MOHD FURQAAN AHMED

Data Engineer | AWS •Azure•Pyspark Databricks • Snowflake • Kafka • Flink

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PROFESSIONAL SUMMARY

- Computer Science graduate with hands-on, self-directed project experience in Data Engineering, focused on SQL, Python, PySpark, cloud data platforms and modern data warehousing.
- Built end-to-end batch ETL/ELT pipelines using **Databricks**, **Snowflake** and **AWS S3**, plus a real-time streaming pipeline using **Kafka**, **Flink** and **Click House**, covering ingestion, transformation, validation, data modeling and analytics-ready datasets.
- Practical, hands-on exposure to **Databricks** (PySpark, Delta Lake), **AWS** (S3), **Snowflake**, **Apache Kafka**, **Apache Flink**, S QL/Jinja, Git/GitHub and Apache Airflow.
- Solid grounding in core data engineering concepts — ETL/ELT, dimensional modeling, incremental processing, partitioning,data quality, joins, CTEs, window functions and performance optimization.
- Comfortable with Agile/Scrum practices, Git-based development and documenting technical workflows; completed self-driven pipeline projects modeled on FMCG, Sports Analytics and Ride-Hailing data scenarios.

TECHNICAL SKILLS

Cloud & Data Platforms: AWS (S3, Glue, IAM, EC2, Redshift), Microsoft Azure (ADF, ADLS, Azure Databricks), Databricks

Data Engineering: ETL/ELT, Batch Processing, Real-Time Stream Processing, Data Pipelines, Data Quality, Incremental Loads,DataModeling

Streaming: Apache Kafka, Debezium (CDC), Apache Flink (Flink SQL), Redis, Schema Registry (Avro), Grafana

Big Data: Apache Spark, PySpark, Spark SQL, Delta Lake, Databricks Notebooks

Databases / Warehouses: Snowflake, ClickHouse, MySQL,

PostgreSQL,Redshift**Programming / Query:** SQL, Python, PySpark, Jinja

Transformation: dbt (Models, Macros, Tests, Seeds, Incremental Models)

Orchestration / Version Control: Apache Airflow, Great Expectations, Git, GitHub

Tools / Methodology: Jupyter Notebook, VS Code / PyCharm, Docker, Jira, Agile/Scrum

PROJECTS

Real-Time CDC and Batch Data Engineering Platform for Retail Analytics

Domain Focus: Retail Analytics

Environment: Python, SQL, PostgreSQL, Apache Kafka, Debezium, Apache Airflow, dbt, Great Expectations, Docker Compose, Git/GitHub

Project Details: A self-driven data platform that combines real-time change data capture (CDC) with scheduled batch ingestion, integrating API, file and database sources into an analytics-ready retail data warehouse.

- Built an end-to-end ELT pipeline integrating REST API, CSV and real-time PostgreSQL CDC data using Debezium, Apache Kafka and Apache Airflow.
- Developed a layered data warehouse architecture (Raw, Staging, Intermediate and Marts) using PostgreSQL and dbt, including incremental transformations.
- Implemented automated data quality checks using Great Expectations and 24 dbt tests to validate data integrity.
- Orchestrated batch and streaming ingestion workflows with Airflow, including SLA monitoring, failure alerts and recovery workflows.
- Generated dbt lineage documentation and implemented source-to-warehouse reconciliation tests for row counts and revenue accuracy.
- Containerized the complete platform using Docker Compose for reproducible local deployment; source code on GitHub (furqaan123456/real-time-cdc-batch-retail-analytics).

Real-Time Ride-Hailing Data Pipeline — Streaming Platform (Kafka + Flink + ClickHouse)

Domain Focus: Ride-Hailing / Mobility

Environment: Python, Apache Kafka, Schema Registry (Avro), Apache Flink (Flink SQL), Redis, ClickHouse, Grafana, DockerCompose,Git/GitHub

Project Details: A self-driven real-time data engineering pipeline that streams ride-hailing events end to end, from eventgeneration to live analytics dashboards.

- Built a Python event simulator publishing ride, driver-location and trip events to Apache Kafka topics with Avro schemas managed through Schema Registry.
- Developed Apache Flink SQL jobs for real-time stream processing, including windowed aggregations on trip anddemandmetrics.
- Loaded processed streams into ClickHouse for low-latency analytical queries, and used Redis as a fast serving layer for live state.
- Created Grafana dashboards for real-time monitoring of key ride-hailing metrics.
- Containerized the full stack with Docker Compose for reproducible local deployment; documented the architecture and setupin the GitHub repository.

Databricks Lakehouse — FMCG Data Engineering Project (Medallion Architecture)

Domain Focus: FMCG (Fast-Moving Consumer Goods)

Environment: Databricks, PySpark, Spark SQL, Delta Lake, Amazon S3, Python, SQL, Jupyter Notebook, Git/GitHub

Project Details: A self-driven end-to-end lakehouse pipeline for an FMCG company that ingests sales and product data, cleans it and publishes analytics-ready datasets for business reporting.

- Ingested raw FMCG sales, customer and product data from Amazon S3 into Databricks using a bronze / silver / gold medallion architecture.

- Used PySpark and Spark SQL notebooks to clean, standardize and transform data, handling duplicates, nulls and data-typeinconsistencies.
- Stored every layer as Delta tables and applied incremental processing to avoid reprocessing unchanged data.
- Modeled fact and dimension tables in the gold layer to support sales and product performance reporting.
- Version-controlled the notebooks in GitHub (project-de-fmcg-atlikon).

Snowflake — Cricket Analytics Data Engineering Project

Domain Focus: Sports Analytics (ODI Cricket)

Environment: Snowflake, SQL, JSON / semi-structured data, Python, Git/GitHub, Dimensional Modeling, ELT

Project Details: A self-driven end-to-end data engineering project on One Day International (ODI) cricket data, taking raw JSON match files through ingestion, profiling, cleansing and transformation to a reporting dashboard in Snowflake.

- Ingested raw ODI cricket match data in JSON format into Snowflake staging tables.
- Performed data profiling to understand structure, null values, duplicates and data quality issues before transformation.
- Cleansed and flattened semi-structured JSON into relational tables using Snowflake SQL.
- Built transformation layers with fact and dimension models for match, team, player and delivery-level analytics.
- Built a data pipeline and an analytics dashboard for metrics such as team performance, player statistics and match results.

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science & Engineering CGPA: 7.9/10
VTU | 2026

TRAINING / PRACTICAL EXPOSURE

- **Data Engineering:** SQL, Python, Git/GitHub, Scrum, Big Data, Apache Airflow and cloud data engineering concepts.
- **Cloud & Lakehouse:** Practical exposure to AWS, Databricks, Delta Lake, cloud storage and data pipeline development.
- **Streaming:** Hands-on practice with Kafka, Flink, ClickHouse and Grafana through a real-time ride-hailing pipeline.
- **Salesforce:** Completed a 90-day Salesforce internship with exposure to Salesforce Administration, Apex, LightningWeb